a *the protocol: a spatial partition, a sandboxed search, and a separated evaluation*

1 Data & partition

Shenzhen · 100 m grid
urban morphology × LST
31 760 cells in scope
A 19 184 · B 6 182
buffer R = 15 (1.5 km) · 3 407
C 2 987 — withheld

2 Discovery-side sandbox

agent reads A + B only
15 candidates · 3 rounds
greedy + local refine
3 151 / 5 000 fits
4 hard physics gates
complexity cap ≤ 15

3 Freeze & hash

4 candidates frozen
SHA-256 canonical hash
best A-CV 2.914 °C
Baseline A fixed (3.158)
placebo = Baseline A
evaluation budget preserved

4 Separated evaluation

C excluded from discovery-side roles
frozen candidate evaluated on C
ΔRMSE +0.3791 °C · p = 0.005
95% CI > 0 · Holm, family = 4
physics audit + negative controls
endpoint passes; placebo fails

5 Protocol audit

post-hoc implementation audit
discovery-side isolation retained
single-access design failed
validation chronology unresolved
Stage 04 reclassified as a
locked-region evaluation (deviation)

the frozen candidates are then stress-tested

b *three adversarial checks on the frozen result*

Conventional learners

RF and GBM, same 7 covariates
out-predict the candidate on C
ΔC +0.7465 vs +0.3791 °C

→ the candidate's advantage is
two-region-specific, not a
transferable law

External replication

Changchun 2021 · 100 647 cells
(post-hoc; 500 m vs Shenzhen 100 m)

partial overlap: nonlinear NDVI
cooling; building warming;
vegetation × building

does not transfer: exp(DEM)
worst; DEM r = -0.07

Constraint & firewall stress test

exploratory combined-pipeline test
unconstrained: A-CV 2.866
(vs H10 2.956), but worse on C
C RMSE 2.625 (vs 2.462)
→ directional, not confirmatory
firewall off manufactures
+0.3503 °C from leakage

the three checks jointly determine the verdict

c *what the protocol returned*

Verdict

No portable urban land-surface-temperature law is established here. The contribution is methodological: an auditable symbolic-search pipeline, a separated discovery-side architecture, and a post-hoc implementation audit showing that autonomous-science workflows must freeze not only data and candidates but also validator software state and access history. What remains is a locked-region evaluation under a documented protocol deviation, plus post-hoc adversarial benchmarking and replication.

The locked-region evaluation is reported under a documented protocol deviation; the Changchun replication and the Stage-05 adversarial benchmarks are post-hoc.